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EARPHONES

Abstract

The present disclosure relates to a loudspeaker assembly comprising a shell assembly, a bone-conducting loudspeaker, an air-conducting loudspeaker, and a circuit system. The shell assembly includes a first accommodation cavity and a second accommodation cavity. The bone-conducting loudspeaker is accommodated in the first accommodation cavity and the air-conducting loudspeaker is accommodated in the second accommodation cavity. The air-conducting loudspeaker includes a magnetic circuit system that includes a magnet and a magnetic conduction assembly. The magnet is fixed on the magnetic conduction assembly, the magnetic conduction assembly is fixed in the second accommodation cavity, and the magnetic conduction assembly is provided with a weight reduction structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Patent Application No. PCT/CN2023/126023, filed on Oct. 23, 2023, which claims priority to Chinese Patent Application No. 202310541798.X, filed on May 12, 2023, the entire contents of each of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of sound-producing apparatus, and in particular, to earphones.

BACKGROUND

[0003] Earphones have been widely used in people's daily lives, and they can be used in conjunction with electronic devices such as smartphones and computers to provide users with an auditory feast. Currently, both air-conducting sound transmission principle and bone-conducting sound transmission principle have been widely applied in earphones. As a result, current earphones are typically either air-conducting type earphones or bone-conducting type earphones, so that the types of headphones are relatively single, which restricts the potential for sound quality improvement. Furthermore, due to the different sound transmission principles of air-conducting loudspeaker and bone-conducting loudspeaker, the requirements for the installation environment and structure differ. Among these, the overall weight of the loudspeaker assembly in the earphone is a major factor influencing the sound quality of the bone-conducting loudspeaker. Therefore, reducing the overall weight of the loudspeaker assembly to enhance the sound quality of earphones that combine both bone-conducting and air-conducting types is an urgent issue that needs to be addressed.

SUMMARY

[0004] The present disclosure provides a loudspeaker assembly. The loudspeaker assembly comprises a shell assembly, a bone-conducting loudspeaker, an air-conducting loudspeaker, and a magnetic circuit system. The shell assembly is provided with a first accommodation cavity and a second accommodation cavity. The bone-conducting loudspeaker is accommodated in the first accommodation cavity and the air-conducting loudspeaker is accommodated in the second accommodation cavity. The air-conducting loudspeaker includes a magnetic circuit system, and the magnetic circuit system includes a magnet and a magnetic conduction assembly, the magnet being fixed on the magnetic conduction assembly, the magnetic conduction assembly being fixed in the second accommodation cavity, and the magnetic conduction assembly being provided with a weight reduction structure.

[0005] In some embodiments, the weight reduction structure includes at least one of a groove or a through hole arranged at the magnetic conduction assembly.

[0006] In some embodiments, a magnetic induction intensity at a position of the magnetic conduction assembly where the weight reduction structure is arranged is less than a magnetic

induction intensity at other positions of the magnetic conduction assembly.

[0007] In some embodiments, the magnetic conduction assembly includes a magnetic conduction shield and a magnetic conduction plate. The magnet is fixed in the magnetic conduction shield and the magnetic conduction plate is fixed to the magnet, the weight reduction structure is arranged on at least one of the magnetic conduction shield or the magnetic conduction plate, there are one or more weight reduction structures on the magnetic conduction shield and the magnetic conduction plate of the magnetic conduction assembly, and a plurality of weight reduction structures are spaced apart.

[0008] In some embodiments, the magnetic conduction shield includes a bottom wall and an annular side wall extending from the bottom wall, and the weight reduction structure is arranged at the bottom wall.

[0009] In some implementations, a total area of the weight reduction structure on the magnetic conduction shield accounts for less than or equal to 50% of an area of a bottom wall of the magnetic conduction shield, and a total area of the weight reduction structure on the magnetic conduction plate accounts for less than or equal to 50% of an area of the magnetic conduction plate.

[0010] In some embodiments, a total area of the weight reduction structure on the magnetic conduction shield accounts for less than or equal to 30% of an area of a bottom wall of the magnetic conduction shield, and a total area of the weight reduction structure on the magnetic conduction plate accounts for less than or equal to 30% of an area of the magnetic conduction plate.

[0011] In some embodiments, a planar shape of the weight reduction structure on the magnetic conduction shield is the same as a planar shape of the bottom wall of the magnetic conduction shield and is reduced in equal proportion relative to the planar shape of the bottom wall of the magnetic conduction shield; and a planar shape of the weight reduction structure on the magnetic conduction plate is the same as a planar shape of the magnetic conduction plate and is reduced in equal proportion relative to the planar shape of the magnetic conduction plate.

[0012] In some embodiments, the weight reduction structure is centrally arranged at the bottom wall.

[0013] In some embodiments, the weight reduction structure is centrally arranged at the magnetic conduction plate.

[0014] In some implementations, the weight reduction structure is a through hole.

[0015] In some embodiments, the magnet is provided with the weight reduction structure, the weight reduction structure is a groove or through hole.

[0016] In some embodiments, a magnetic induction intensity at a position of the magnet where the weight reduction structure is arranged is less than a magnetic induction intensity at other positions of the magnet.

[0017] In some embodiments, the weight reduction structure on the magnet is located at a center of the magnet, and a planar shape of the weight reduction structure on the magnet is the same as a planar shape of the magnet and is reduced in equal proportion relative to the planar shape of the magnet.

[0018] In some embodiments, a planar area of the weight reduction structure on the magnet accounts for less than or equal to 50% of a planar area of the magnet.

[0019] In some implementations, the air-conducting loudspeaker further includes a fixing assembly, the fixing assembly is fixed to the magnetic conduction assembly, and the fixing assembly is fixedly connected to the shell assembly.

[0020] In some embodiments, the air-conducting loudspeaker further includes a vibration diaphragm and a voice coil, a rim of the vibration diaphragm is connected to a periphery of the fixing assembly, and the voice coil is connected to the vibration diaphragm.

[0021] In some embodiments, the loudspeaker assembly is provided with a pressure relief hole for communicating with the second accommodation cavity and external environment, and the pressure relief hole extends toward a side where the first accommodation cavity is located.

[0022] The present disclosure provides a wearable electronic device, comprising the loudspeaker assembly of any one of the above embodiments.

[0023] In some embodiments, the wearable electronic device wearable electronic device is earphones, glasses, an AR device, or a VR device.

[0024] The beneficial effect of the present disclosure is as follows: with such an arrangement, the overall weight of the air-conducting loudspeaker can be effectively reduced, and accordingly, effectively reducing the overall weight of the loudspeaker assembly, improving the user's experience, and effectively improving the operational stability of the bone-conducting loudspeaker, and improving the sound transmission quality of the bone-conducting loudspeaker, thereby effectively improve the sound quality of the earphone.

[0025] It should be appreciated that the above general description and the detailed descriptions that follow are exemplary and explanatory only, and are not intended to limit the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] In order to more clearly illustrate the technical solutions in the embodiments of the present disclosure, the accompanying drawings that need to be used in the description of the embodiments will be briefly introduced in the following, and it will be obvious that the accompanying drawings in the following description are only some of the embodiments of the present disclosure, and other drawings can be obtained based on them without creative labor to a person of ordinary skill in the art.

[0027] FIG. 1 is a schematic diagram illustrating a three-dimensional structure of an earphone according to embodiments of the present disclosure;

[0028] FIG. 2 is a schematic diagram illustrating a three-dimensional structure of a loudspeaker assembly and a wearing assembly shown in FIG. 1;

[0029] FIG. 3 is an exploded schematic diagram of a structure of the loudspeaker assembly shown in FIG. 2;

[0030] FIG. 4 is an exploded schematic diagram of a structure of an air-conducting loudspeaker shown in FIG. 2;

[0031] FIG. 5 is a schematic diagram illustrating a cross-section of a structure of the air-conducting loudspeaker shown in FIG. 2;

[0032] FIG. 6 is a schematic diagram illustrating a three-dimensional structure of a magnetic conduction shield shown in FIG. 4;

[0033] FIG. 7 is a schematic diagram illustrating a three-dimensional structure of a magnetic conduction plate shown in FIG. 4;

[0034] FIG. 8 is a line graph illustrating a relationship curve between a preset proportion of a total area of a weight reduction structure on the magnetic conduction shield in FIG. 6 to a total area of the magnetic conduction shield and a weight change of the loudspeaker assembly, and a relationship curve between the preset proportion and a sensitivity change of the air-conducting loudspeaker.

[0035] FIG. 9 is a schematic diagram illustrating a three-dimensional structure of a magnet shown in FIG. 4;

[0036] FIG. 10 is an exploded schematic diagram of a first embodiment of a pressure relief hole protection scheme of the loudspeaker assembly shown in FIG. 2;

[0037] FIG. 11 is a schematic diagram illustrating a partially assembled structure of the first embodiment of the pressure relief hole protection scheme shown in FIG. 10;

[0038] FIG. 12 is a schematic diagram illustrating a three-dimensional structure of a protective mesh shown in FIG. 10;

[0039] FIG. **13** is a schematic diagram illustrating a front view of a second embodiment of the pressure relief hole protection scheme of the loudspeaker assembly shown in FIG. **2**;

[0040] FIG. **14** is a schematic diagram illustrating a sectional structure of an embodiment of a region A shown in FIG. **13**; and

[0041] FIG. **15** is a schematic diagram illustrating a sectional structure of another embodiment of the region A shown in FIG. **13**.

DETAILED DESCRIPTION

[0042] In order to enable those skilled in the art to better understand the technical solutions of the present disclosure, the present disclosure is described in further detail below in conjunction with the accompanying drawings and specific embodiments. It is to be understood that the described embodiments are only a part of the embodiments of the present disclosure, and not all of them. Based on the embodiments in the present disclosure, all other embodiments obtained by a person of ordinary skill in the art without making creative labor fall within the scope of protection of the present disclosure.

[0043] The terms “first”, “second”, etc. in the present disclosure are used to differentiate between different objects and are not used to describe a particular order. Additionally, the terms “comprising” and “including”, and any variations thereof, are intended to cover non-exclusive inclusion. For example, a process, method, system, product, or apparatus comprising a series of steps or units is not limited to the listed steps or units, but optionally also includes steps or units that are not listed, or optionally also includes other steps or units inherent to those processes, methods, products, or devices.

[0044] The present disclosure proposes a wearable electronic device. The wearable electronic device includes a movement assembly. The wearable electronic device includes earphones, eyeglasses, an augmented reality (AR) device, or a virtual reality (VR) device. The following describes an exemplary structure of the wearable electronic device through a practical application of the wearable electronic device of the present disclosure in earphones.

[0045] As shown in FIG. **1**, the earphone **100** may include loudspeaker assemblies **1**, ear-hook assemblies **2**, and a rear-hook assembly **3**. There may be two loudspeaker assemblies **1**. The two loudspeaker assemblies **1** are configured to transmit vibration and/or sound to the user's left ear and right ear, respectively. The two loudspeaker assemblies **1** may be the same or different. For example, one loudspeaker assembly **1** may be provided with a microphone, and the other loudspeaker assembly **1** may not be provided with a microphone. As another example, one loudspeaker assembly **1** may be provided with a key and a corresponding circuit board, and another loudspeaker assembly **1** may not be provided with the key and the corresponding circuit board. The two loudspeaker assemblies **1** may be of the same configuration in terms of movement modules (e.g., loudspeaker modules). The loudspeaker assembly **1** described hereinafter may be considered to be described in detail by taking one of the two loudspeaker assemblies **1** as an example. There may be two ear-hook assemblies **2**, and the two ear-hook assemblies **2** may be located in the user's left and right ears, respectively, so that the loudspeaker assemblies **1** can fit the user's face. For example, one of the ear-hook assemblies **2** may be provided with a battery, and another ear-hook assembly **2** may be provided with a control circuit, or the like. One end of an ear-hook assembly **2** is connected to a loudspeaker assembly **1**, and another end of the ear-hook assembly **2** is connected to the rear-hook assembly **3**. The rear-hook assembly **3** connects the two ear-hook assemblies **2**, and the rear-hook assembly **3** is configured to be wrapped around the back of the user's neck or the back of the user's brain and provides a clamping force, which allows the two loudspeaker assemblies **1** to be clamped to both sides of the user's face or firmly arranged on the outside of the user's ears and allows the ear-hook assembly **2** to be more securely hang on the user's ears. It should be understood that the earphone **100** may not include the rear-hook assembly **3**, with the loudspeaker assemblies **1** being worn in the user's ears through the ear-hook assemblies **2**. Alternatively, the earphone **100** may not include the rear-hook assembly **3** and the ear-hook

assemblies **2**, and the loudspeaker assemblies **1** are connected by a headband structure or a neck-hook structure, the loudspeaker assemblies **1** are attached to the user's ears or securely disposed on the outside of the user's ears through the headband structure or the neck-hook structure.

[0046] The following primarily provides an exemplary description of a loudspeaker assembly **1** of the earphone **100**. In some embodiments, the loudspeaker assembly **1** may also be referred to as a movement assembly.

[0047] Optionally, as shown in FIG. **1**, FIG. **2**, FIG. **3**, FIG. **4**, and FIG. **5**, the loudspeaker assembly **1** includes a shell assembly **10**, a bone-conducting loudspeaker **11**, and an air-conducting loudspeaker **12**.

[0048] The shell assembly **10** is provided with a first accommodation cavity **1001** and a second accommodation cavity **1002**. The bone-conducting loudspeaker **11** is accommodated in the first accommodation cavity **1001**, and the air-conducting loudspeaker **12** is accommodated in the second accommodation cavity **1002**. The air-conducting loudspeaker **12** includes a magnetic circuit system **1260**. The magnetic circuit system **1260** includes a magnet **1210** and a magnetic conduction assembly **1220**. The magnet **1210** is fixed to the magnetic conduction assembly **1220**, the magnetic conduction assembly **1220** is fixed in the second accommodation cavity **1002**, and the magnetic conduction assembly **1220** is provided with a weight reduction structure **1223**.

[0049] Specifically, in one embodiment, the bone-conducting loudspeaker **11** and the air-conducting loudspeaker **12** are simultaneously provided in the same shell assembly **10**, and the bone-conducting loudspeaker **11** drives the loudspeaker assembly **1** to vibrate as a whole when it is in operation to transmit the sound to the user's ear. The air-conducting loudspeaker **12** includes the magnetic circuit system **1260**. The magnetic circuit system **1260** includes the magnet **1210** and the magnetic conduction assembly **1220**. Further, the magnetic conduction assembly **1220** is provided with the weight reduction structure **1223**. The weight reduction structure **1223** is configured to reduce the overall weight of the air-conducting loudspeaker **12**, thereby effectively reducing the overall weight of the loudspeaker assembly **1**, improving the user experience, and effectively enhancing the operational stability of the bone-conducting loudspeaker **11**. Moreover, as the bone-conducting loudspeaker **11** needs to drive the overall vibration of the loudspeaker assembly **1** to transmit sound to the user's ear when the bone-conducting loudspeaker **11** is working, decreasing the overall weight of the loudspeaker assembly **1** can improve the sensitivity of the bone-conducting loudspeaker **11**, thereby effectively improving the sound transmission quality of the bone-conducting loudspeaker **11** and effectively improving the sound quality of the earphone.

[0050] Optionally, the weight reduction structure **1223** includes at least one of a groove or a through hole arranged at the magnetic conduction assembly **1220**. In some embodiments, a through hole or a groove may be arranged at the magnetic conduction assembly **1220** to serve as the weight reduction structure **1223** of the magnetic conduction assembly **1220**, or both a through hole and a groove may be arranged at the magnetic conduction assembly **1220** to serve as the weight reduction structure **1223** of the magnetic conduction assembly **1220**. For example, in one embodiment, the weight reduction structure **1223** is realized by arranging a through hole at the magnetic conduction assembly **1220**, which effectively reduces the overall weight of the loudspeaker assembly **1**.

[0051] Optionally, a magnetic induction intensity at a position of the magnetic conduction assembly **1220** where the weight reduction structure **1223** is arranged is less than a magnetic induction intensity at other positions of the magnetic conduction assembly **1220**. By arranging the groove and/or the through hole at a position of the magnetic conduction assembly **1220** where the magnetic induction intensity is smaller, it is possible to minimize the influence of arranging the groove or the through hole at the magnetic conduction assembly **1220** on the magnetic conduction function of the magnetic conduction assembly **1220**, and ensure that the overall weight of the loudspeaker assembly **1** can be effectively reduced while affecting the magnetic conduction function of the magnetic conduction assembly **1220** as little as possible.

[0052] Optionally, the magnetic conduction assembly **1220** includes a magnetic conduction shield

1221 and a magnetic conduction plate **1222**. The magnet **1210** is fixed in the magnetic conduction shield **1221**, and the magnetic conduction plate **1222** is fixed to the magnet **1210**. The weight reduction structure **1223** is arranged on at least one of the magnetic conduction shield **1221** or the magnetic conduction plate **1222**, and there may be one or more weight reduction structures **1223** on the magnetic conduction shield **1221** and the magnetic conduction plate **1222**. When a plurality of weight reduction structures **1223** are employed, the plurality of weight reduction structures **1223** are spaced apart.

[0053] Specifically, the magnetic conduction shield **1221** is provided with a magnet accommodation slot for accommodating and fixing the magnet **1210**. The magnet **1210** is fixed in the magnet accommodation slot, and the magnetic conduction plate **1222** is fixed to a side of the magnet **1210** away from the magnet accommodation slot. For example, referring to FIG. 5, in one embodiment, the magnetic conduction shield **1221** and the magnetic conduction plate **1222** are provided with one or more weight reduction structures **1223**. In other embodiments, one or more weight reduction structures **1223** may be arranged on only the magnetic conduction plate **1222** or only magnetic conduction shield **1221**. For example, in one embodiment, one weight reduction structure **1223** is arranged on the magnetic conduction shield **1221**, and one weight reduction structure **1223** is correspondingly arranged on the magnetic conduction plate **1222**. Based on this, the overall weight of the loudspeaker assembly **1** can be further reduced, thereby effectively enhancing the sound quality of the earphone.

[0054] Optionally, in one embodiment, the magnetic conduction shield **1221**, the magnet **1210**, and the magnetic conduction plate **1222** are sequentially centered along an air-conducting operating direction **X1** of the air-conducting loudspeaker **12**, which effectively ensures the centering of the center of mass of the magnetic conduction assembly **1220**. The air-conducting operating direction **X1** refers to a vibration direction of a vibration diaphragm of the air-conducting loudspeaker **12** during operation.

[0055] Optionally, as shown in FIG. 6, the magnetic conduction shield **1221** includes a bottom wall **1224** and an annular side wall **1225** extending from the bottom wall **1224**, and a weight reduction structure **1223b** is arranged at the bottom wall **1224**. Specifically, referring to FIG. 6, in one embodiment, the magnetic conduction shield **1221** includes the bottom wall **1224** and the annular side wall **1225** extending from the bottom wall **1224**. The bottom wall **1224** and the annular side wall **1225** enclose to form the magnet accommodation slot. Further, in one embodiment, the weight reduction structure **1223b** is arranged at the bottom wall **1224**. For example, a groove or a through hole, etc., is arranged at the bottom wall **1224**. Based on this, the amount of material used for the bottom wall **1224** can be effectively reduced, which effectively reduces the weight of the magnetic conduction shield **1221** and reduces the weight of the air-conducting loudspeaker **12**, thereby improving the operational stability of the bone-conducting loudspeaker **11** and improving the sound quality of the earphone.

[0056] Optionally, refer to FIG. 5 and FIG. 6, the weight reduction structure **1223b** is centrally arranged at the bottom wall **1224**. Specifically, the weight reduction structure **1223b** on the magnetic conduction shield **1221** is centrally arranged at the bottom wall **1224** of the magnetic conduction shield **1221**, and a magnetic induction intensity at the middle of the bottom wall **1224** of the magnetic conduction shield **1221** is smaller compared to other positions of the magnetic conduction shield **1221**. Therefore, the weight reduction structure **1223b** being centrally arranged at the bottom wall **1224** can effectively reduce the influence of the weight reduction structure **1223b** on the magnetic conduction function of the magnetic conduction assembly **1220** and can effectively ensure that the center of mass of the magnetic conduction shield **1221** is centrally aligned, thereby effectively enhancing the operational stability of the bone-conducting loudspeaker **11** and enhancing the sound quality of the earphone.

[0057] It will be appreciated that in some other embodiments of the present disclosure, the weight reduction structure **1223b** may also be arranged at other positions of the bottom wall **1224** of the

magnetic conduction shield **1221**.

[0058] Optionally, refer to FIG. 5 and FIG. 7, the weight reduction structure **1223a** is centrally arranged at the magnetic conduction plate **1222**. Specifically, the weight reduction structure **1223a** is centrally arranged at the magnetic conduction plate **1222**. The center of mass of components of the air-conducting loudspeaker **12** such as the magnetic conduction plate **1222**, the magnetic conduction shield **1221**, the magnet **1210** is centrally arranged and positioned at the same line along the air-conducting operating direction **X1**. Based on this, the center of mass of the air-conducting loudspeaker **12** can be ensured to be approximately located at the geometric center of the air-conducting loudspeaker **12**. Centrally arranging the weight reduction structure **1223a** can effectively prevent the weight reduction structure **1223a** from affecting the center of mass of the magnetic conduction plate **1222** to cause a displacement of the center of mass of the air-conducting loudspeaker **12**, thereby enhancing the operational stability of the bone-conducting loudspeaker **11** and improving the sound quality of the earphone. Moreover, a magnetic induction intensity at the middle of the magnetic conduction plate **1222** is smaller compared to a magnetic induction intensity at the other positions of the magnetic conduction plate, therefore, the weight reduction structure **1223a** centrally arranging at the magnetic conduction plate **1222** can reduce the weight of the air-conducting loudspeaker **12** while reducing the influence of the weight reduction structure **1223a** on the magnetic conduction function of the magnetic conduction assembly **1220**.

[0059] Optionally, a total area of the weight reduction structure on the magnetic conduction shield **1221** accounts for less than or equal to a preset proportion of an area of the bottom wall **1224** of the magnetic conduction shield **1221**; a total area of the weight reduction structure on the magnetic conduction plate **1222** accounts for less than or equal to a preset proportion of an area of the magnetic conduction plate **1222**. Within the preset proportion, a magnetic induction intensity at a position where the weight reduction structure is located is smaller compared to other positions, thereby reducing the weight of the air-conducting loudspeaker **12** while reducing the influence of the weight reduction structure **1223a** on the magnetic conduction function of the magnetic conduction assembly **1220**.

[0060] Specifically, referring to FIG. 5, FIG. 6, and FIG. 7, in one embodiment, both the magnetic conduction plate **1222** and the magnetic conduction shield **1221** are arranged with a weight reduction structure. The area of the bottom wall **1224** of the magnetic conduction shield **1221** is an area enclosed by an outer rim contour **1224a** of the bottom wall **1224**, and the total area of the weight reduction structure **1223b** on the magnetic conduction shield **1221** is an area of a region enclosed by an outer rim contour **1224b** of the weight reduction structure **1223b**. In one embodiment, the preset proportion is 50%. In other words, the total area of the weight reduction structure **1223b** on the magnetic conduction shield **1221** accounts for less than or equal to 50% of the area of the bottom wall **1224** of the magnetic conduction shield **1221**. Based on this, it can effectively reduce the weight of the magnetic conduction shield **1221** without affecting the magnetic conduction function of the magnetic conduction shield **1221**. Similarly, the area of the magnetic conduction plate **1222** is an area enclosed by an outer rim contour **1222a** of the magnetic conduction plate **1222**, and the total area of the weight reduction structure **1223a** on the magnetic conduction plate **1222** is the total area enclosed by an outer rim contour **1222b** of the weight reduction structure. The total area of the weight reduction structure **1223a** on the magnetic conduction plate **1222** accounts for less than or equal to 50% of the area of the magnetic conduction plate **1222**. Based on this, it can efficiently reduce the weight of the magnetic conduction plate **1222** without affecting the magnetic conduction function of the magnetic conduction plate **1222**. In some embodiments, if a plurality of weight reduction structures **1223b** are arranged on the magnetic conduction shield **1221**, the total area of the weight reduction structures **1223b** on the magnetic conduction shield **1221** is the sum of the areas of the plurality of weight reduction structures **1223b**. Similarly, if a plurality of weight reduction structures **1223a** are arranged on the magnetic conduction plate **1222**, the total area of the weight reduction structures on

the magnetic conduction plate **1222** is the sum of the areas of the plurality of weight reduction structures **1223b**.

[0061] As shown in FIG. **8**, taking the preset proportion of the total area of the weight reduction structure **1223b** of the magnetic conduction shield **1221** to the total area of the bottom wall **1224** as an example, a simulation experiment was conducted to obtain a relationship curve **Z2** between the preset proportion and a sensitivity change value of the air-conducting loudspeaker **12** and a relationship curve **Z1** between the preset proportion and a weight change value of the loudspeaker assembly **1**. Referring to the relationship curve **Z1**, the weight change value of the loudspeaker assembly **1** increases as the preset proportion increases and the thickness of the bottom wall **1224** reduces, i.e., the weight of the loudspeaker assembly **1** decreases as the preset proportion increases and the thickness of the bottom wall **1224** decreases. Referring to the relationship curve **Z2**, as the preset proportion increases, the sensitivity change value of the air-conducting loudspeaker **12** tends to increase slightly and then decrease when the thickness of the bottom wall **1224** is the same. Therefore, selecting a preset proportion of less than or equal to 50% can effectively ensure the weight reduction influence of the loudspeaker assembly **1** while reducing the influence of the weight reduction structure **1223b** on the magnetic conduction function of the magnetic conduction shield **1221**. Moreover, it can be seen that when the thickness of the bottom wall **1224** is the same, the preset proportion of less than or equal to 30% can reduce the weight of the loudspeaker assembly **1** with less influence on the magnetic conduction function of the magnetic conduction shield **1221**. The magnetic conduction plate **1222** may also undergo similar simulation experiments to obtain the corresponding experimental data. It can be concluded that when the preset proportion is less than or equal to 30%, the loudspeaker assembly **1** can achieve a better weight reduction effect while minimizing the influence on the magnetic conduction function of the magnetic conduction plate **1222**. The magnet **1210** hereinafter is similar, and will not be repeated herein in detail.

[0062] Optionally, in some embodiments, the preset proportion may also be 50% and greater than 0. For example, the preset proportion may be 45%, 40%, 35%, 28%, 20%, 10%, 5%, or the like. In other words, the total area of the weight reduction structure **1223b** on the magnetic conduction shield **1221** accounts for less than or equal to 50% of the area of the bottom wall **1224** of the magnetic conduction shield **1221**. Based on this, the weight of the magnetic conduction shield **1221** can be effectively reduced without affecting the magnetic conduction function of the magnetic conduction shield **1221**. The total area of the weight reduction structure **1223a** on the magnetic conduction plate **1222** accounting for less than 50% of the area of the magnetic conduction plate **1222** can effectively reduce the weight of the magnetic conduction plate **1222** without affecting the magnetic conduction function of the magnetic conduction plate **1222**.

[0063] In some embodiments, only one of the magnetic conduction plate **1222** or the magnetic conduction shield **1221** is arranged with the weight reduction structure **1223**. A proportion of the total area of the weight reduction structure **1223** to the area of the magnetic conduction plate **1222** or the bottom wall **1224** of the magnetic conduction shield **1221** can be seen referred to in the descriptions above, and will not be repeated herein in detail.

[0064] Optionally, referring to FIG. **6**, a planar shape of the weight reduction structure **1223b** is the same as a planar shape of the bottom wall **1224** of the magnetic conduction shield **1221** and is reduced in equal proportion relative to the planar shape of the bottom wall **1224** of the magnetic conduction shield **1221**. Specifically, the planar shape of the weight reduction structure **1223b** is the same as the planar shape of the bottom wall **1224** of the magnetic conduction shield **1221** and is reduced in equal proportion relative to the planar shape of the bottom wall **1224** of the magnetic conduction shield **1221**, it may be understood that the planar shape of the weight reduction structure **1223b** on the magnetic conduction shield **1221** perpendicular to the air-conducting operation direction **X1** is the same as the planar shape of the bottom wall of the magnetic conduction shield **1221** perpendicular to the air-conducting operation direction **X1**, and is reduced

in equal proportion relative to the planar shape of the bottom wall of the magnetic conduction shield **1221**. Based on this, it can effectively ensure the alignment of the center of mass of the magnetic conduction shield **1221** and a center of the magnetic conduction shield **1221**. It is to be understood that in other embodiments of the present disclosure, the planar shape of the weight reduction structure **1223b** on the magnetic conduction shield **1221** perpendicular to the air-conducting operating direction **X1** and the planar shape of the bottom wall of the magnetic conduction shield **1221** perpendicular to the air-conducting operating direction **X1** may also be different.

[0065] Optionally, referring further to FIG. 7, the planar shape of the weight reduction structure **1223a** on the magnetic conduction plate **1222** perpendicular to the air-conducting operating direction **X1** is the same as the planar shape of the magnetic conduction plate **1222** perpendicular to the air-conducting operating direction **X1**, and is reduced in equal proportion relative to the planar shape of the magnetic conduction plate **1222**. Based on this, it can effectively ensure the alignment of the center of mass of the magnetic conduction plate **1222** and a center of the magnetic conduction plate **1222**. For example, in one embodiment, a weight reduction structure **1223b** with a planar shape that is the same as the planar shape of the magnetic conduction shield **1221** and is reduced in equal proportion relative to the planar shape of the magnetic conduction shield **1221** is centrally arranged on the magnetic conduction shield **1221**, and a weight reduction structure **1223b** with a planar shape that is the same as the planar shape of the magnetic conduction plate **1222** and is reduced in equal proportion relative to the planar shape of the magnetic conduction plate **1222** is centrally arranged on the magnetic conduction plate **1222**. Based on this, it can effectively ensure the alignment of the center of mass of the air-conducting loudspeaker **12** and a center of the air-conducting loudspeaker **12**, and reduce the weight of the loudspeaker assembly **1**, thereby effectively improving the sound quality of the earphone. It is to be understood that, in other embodiments of the present disclosure, the planar shape of the weight reduction structure **1223a** on the magnetic conduction plate **1222** perpendicular to the air-conducting operating direction **X1** and the planar shape of the magnetic conduction plate **1222** perpendicular to the air-conducting operating direction **X1** may also be different.

[0066] Optionally, referring to FIG. 5 and FIG. 9, the magnet **1210** may be arranged with a weight reduction structure **1223c**, and the weight reduction structure **1223c** is a groove or a through hole. In one embodiment, the weight reduction structure **1223c** is a through hole. Based on this, the weight of the loudspeaker assembly **1** can be further reduced, thereby effectively enhancing the sound quality of the earphone.

[0067] Optionally, a magnetic induction intensity at a position of the magnet **1210** where the weight reduction structure **1223c** is arranged is less than a magnetic induction intensity at other positions of the magnet **1210**. Specifically, the magnetic induction intensity at the position of the magnet **1210** where the weight reduction structure **1223c** is arranged is less than a magnetic induction intensity at other positions of the magnet **1210**. This setting can effectively reduce the overall weight of the loudspeaker assembly **1** without affecting the magnetic conduction function of the magnetic conduction assembly **1220**.

[0068] Optionally, referring further to FIG. 5 and FIG. 9, the weight reduction structure **1223c** on the magnet **1210** is located at a center of the magnet **1210**, and a planar shape of the weight reduction structure **1223c** on the magnet **1210** is the same as a planar shape of the magnet **1210** and is reduced in equal proportion relative to the planar shape of the magnet **1210**. Specifically, the planar shape of the weight reduction structure **1223c** on the magnet **1210** is the same as the planar shape of the magnet **1210** and is reduced in equal proportion relative to the planar shape of the magnet **1210**, and it may be understood that the planar shape of the weight reduction structure **1223c** on the magnet **1210** perpendicular to the air-conducting operating direction **X1** is the same as the planar shape of the magnet **1210** perpendicular to the air-conducting operating direction **X1** and is reduced in equal proportion relative to the planar shape of the magnet **1210**. The weight

reduction structure **1223c** is arranged on the magnet **1210** based on the above-described manner, which can effectively ensure the alignment of the center of mass of the air-conducting loudspeaker **12** and a center of the air-conducting loudspeaker **12** while effectively reducing the weight of the loudspeaker assembly **1**, thus effectively enhancing the sound quality of the earphone.

[0069] Optionally, a planar area of the weight reduction structure **1223c** on the magnet **1210** is less than a preset proportion of the planar area of the magnet **1210**. In one embodiment, the preset proportion is 50%, and in other embodiments, the preset proportion may be other values. The planar area of the weight reduction structure **1223c** is an area enclosed by an outer rim contour **1210b** of the weight reduction structure **1223c**, and the planar area of the magnet **1210** is an area enclosed by an outer rim contour **1210a** of the magnet **1210**. Based on this, it can effectively reduce the weight of the magnet **1210** without affecting the magnetic conduction function of the magnet **1210**.

[0070] Optionally, the weight reduction structure **1223** does not need to be filled with other materials, such as a plastic member, that is, the weight reduction structure **1223** remains in a hollow state. In this way, the weight reduction effect can be fully achieved.

[0071] Optionally, referring to FIG. 4 and FIG. 5, the air-conducting loudspeaker **12** further includes a fixing assembly **1230**. The fixing assembly **1230** is fixed to the magnetic conduction assembly **1220** and is fixedly connected to the shell assembly **10**. The magnetic conduction assembly **1220** arranged with the magnet **1210** is fixedly connected to the shell assembly **10** through the fixing assembly **1230**, which facilitates fixation of the magnetic conduction assembly **1220** in the shell assembly **10**, enhances the connection stability of the magnetic conduction assembly **1220** and the shell assembly **10**, and reduces the jitter of the magnetic conduction assembly **1220** relative to the shell assembly **10**, thereby effectively improve the operational stability of the air-conducting loudspeaker **12** and improving the sound quality of the earphone.

[0072] Optionally, referring to FIG. 4 and FIG. 5, the air-conducting loudspeaker **12** further includes a vibration diaphragm **1240** and a voice coil **1250**, with a rim of the vibration diaphragm **1240** being fixed to a periphery of the fixing assembly **1230**, and the voice coil **1250** being fixed to the vibration diaphragm **1240**. Specifically, in one embodiment, the voice coil **1250** is around an outer peripheral side of the magnet **1210** and the magnetic conduction plate **1222**, the rim of the vibration diaphragm **1240** is connected to the periphery of the fixing assembly **1230**, and the voice coil **1250** is connected to the vibration diaphragm **1240**. For example, as shown in FIG. 4 and FIG. 5, in one embodiment, the fixing assembly **1230** includes a fixing ring **1231** and a fixing body **1232**. The fixing ring **1231** is fitted onto an outer circumference of the annular side wall **1225** of the magnetic conduction shield **1221** and is located between an inner sidewall of the fixing body **1232** and the outer circumference of the annular side wall **1225**. Based on this, the fixing ring **1231** is configured to provide a tension force. The rim of the vibration diaphragm **1240** is connected to the periphery of the fixing body **1232**. When assembling the fixing assembly **1230** and the magnetic conduction shield **1221**, after the fixing ring **1231** is fitted onto the outer circumference of the annular side wall **1225** of the magnetic conduction shield **1221**, the magnetic conduction shield **1221** fitted with the fixing ring **1231** may be pushed into an inner circumference of the fixing body **1232** along the air-conducting operating direction **X1**, which quickly fixes the magnetic conduction shield **1221** to the fixing assembly **1230**, thereby enabling the magnetic conduction shield **1221** to maintain a stable and fixed connection with the fixing body **1232** while effectively improving the assembly efficiency of the air-conducting loudspeaker **12**.

[0073] Optionally, referring to FIG. 10 and FIG. 11, the shell assembly **10** is provided with a pressure relief hole **1008** for communicating the second accommodation cavity **1002** and the external environment, and the pressure relief hole **1008** extends toward a side where the first accommodation cavity **1001** is located. The shell assembly **10** is provided with the pressure relief hole **1008** for relieving pressure for the air-conducting loudspeaker **12**, the pressure relief hole **1008** communicates with the second accommodation cavity **1002** and the external environment,

and the pressure relief hole **1008** extends toward the side where the first accommodation cavity **1001** is located from the communication position between the second accommodation cavity **1002** and the pressure relief hole **1008**. For example, a portion of a hole wall of the pressure relief hole **1008** extends from the communication position of the second accommodation cavity **1002** and the pressure relief hole **1008** toward the side where the first accommodation cavity **1001** is located, which can effectively reduce the occupation of the outer shell space of the first accommodation cavity **1001** by the pressure relief hole **1008** while efficiently utilizing the space between the first accommodation cavity **1001** and the second accommodation cavity **1002** and enhancing the spatial utilization of the shell assembly **10**, allowing for a more compact and reasonable design of the shell assembly **10** while still meeting the acoustic requirements of the air-conducting loudspeaker **12**. Besides, such an inclined extension can increase the dimension of the pressure relief hole **1008**, which makes it possible to provide the air-conducting loudspeaker **12** with better sound quality. [0074] Optionally, referring to FIG. 2 and FIG. 3, the shell assembly **10** may include a shell **101**, a shell **102**, and a shell **103**. The shell **101** and the shell **102** fit with each other to form the first accommodation cavity **1001**. The shell **101** and/or the shell **102** may form a portion of the second accommodation cavity **1002**, and the shell **103** may form the other portion of the second accommodation cavity **1002**, so that the shell **103** and the shell **101** and/or the shell **102** fits with each other to form the second accommodation cavity **1002**.

[0075] The shell assembly **10** may be composed of the shell **101**, the shell **102**, and the shell **103** fitting with each other. The shell **101** and the shell **102** fit with each other to form the first accommodation cavity **1001**, and the shell **103** and the shell **101** fit with each other to form the second accommodation cavity **1002**. The shell **101**, the shell **102**, and the shell **103** fitting with each other to form the shell assembly **10** can make the structure of the loudspeaker assembly **1** compact while facilitating the assembly of the loudspeaker assembly **1**, to enhance the assembly efficiency of the loudspeaker assembly **1**. In some embodiments, the shell **101** may be referred to as a first shell; the shell **102** may be referred to as a second shell; and the shell **103** may be referred to as a third shell.

[0076] The shell assembly **10** may include the shell **101**, the shell **102**, and the shell **103**. A partition wall **1012** may be disposed on the shell **101** and/or the shell **102**, the shell **101** and the shell **102** fitting with each other to form the first accommodation cavity **1001** and the pressure relief hole **1008**. For example, the shell **101** and/or the shell **102** further form a portion of the second accommodation cavity **1002**. For example, the shell **103** may form the other portion of the second accommodation cavity **1002**, with the shell **103** fitting with the shell **101** and/or the shell **102** to form the second accommodation cavity **1002**. Disposing the partition wall **1012** to the shell **101** and/or the shell **102** may be understood as the partition wall **1012** is a part of the shell **101** and/or the shell **102**. It should be understood that how the partition wall **1012** is disposed is not limited to being disposed on the shell **101** and/or the shell **102**. In other embodiments, the partition wall **1012** may be a component that is independent of the shell **101** and/or the shell **102**.

[0077] As shown in FIG. 3, the partition wall **1012** is disposed on the shell **101**, and the shell **101** includes a sub-accommodation cavity **1010** and a sub-accommodation cavity **1011** located on two opposite sides of the partition wall **1012**, and an opening direction of the sub-accommodation cavity **1010** is along a wall surface of the partition wall **1012**. Specifically, the opening direction of the sub-accommodation cavity **1010** is parallel or substantially parallel to the wall surface of the partition wall **1012**. The opening direction of the sub-accommodation cavity **1011** is configured as to intersect with the wall surface of the partition wall **1012**. The shell **102** is provided with a sub-accommodation cavity **1020**, the shell **102** is capped at an opening end of the sub-accommodation cavity **1010**, and the sub-accommodation cavity **1020** and the sub-accommodation cavity **1010** fit with each other to form the sub-accommodation cavity **1010**. The shell **103** is provided with a sub-accommodation cavity **1030**, the shell **103** is capped at an opening end of the sub-accommodation cavity **1011**, and the sub-accommodation cavity **1030** fits with the sub-accommodation cavity **1011**

to form the second accommodation cavity **1002**. In some embodiments, the sub-accommodation cavity **1010** may also be referred to as a first sub-accommodation cavity; the sub-accommodation cavity **1011** may also be referred to as a second sub-accommodation cavity; the sub-accommodation cavity **1020** may also be referred to as a third sub-accommodation cavity; the sub-accommodation cavity **1030** may also be referred to as a fourth sub-accommodation cavity.

[0078] By disposing the partition wall **1012**, the shell **101** is divided into the sub-accommodation cavity **1010** whose opening direction is along the wall surface of the partition wall **1012** and the sub-accommodation cavity **1011** whose opening direction intersects with crosswise with the wall surface of the partition wall **1012**, so that a large space can be realized for both the sub-accommodation cavity **1010** and the sub-accommodation cavity **1011**, improving the space utilization rate of the shell **101** and reducing the mutual interference between the bone-conducting loudspeaker **11** and the air-conducting loudspeaker **12** vibrating with each other in operation, thereby improving the sound quality.

[0079] Optionally, as shown in FIG. 3, FIG. 10, FIG. 11, and FIG. 12, the present disclosure also proposes a first embodiment of a pressure relief hole **1008** protection scheme for protecting the pressure relief hole **1008**. The first embodiment of the pressure relief hole **1008** protection scheme is applicable to the earphone provided with both the bone-conducting loudspeaker **11** and the air-conducting loudspeaker **12**, and may also be applicable to the earphone **100** provided with the air-conducting loudspeaker **12** alone. Herein, the first embodiment of the pressure relief hole **1008** protection scheme is described in terms of the earphone **100** provided with both the air-conducting loudspeaker **12** and the bone-conducting loudspeaker **11**. The first embodiment of the pressure relief hole **1008** protection scheme is described as follows.

[0080] The loudspeaker assembly **1** includes a protective mesh **13**. The shell assembly **10** is provided with an accommodation cavity and the pressure relief hole **1008** for communicating the accommodation cavity and the external environment. An opening mouth of the pressure relief hole **1008** is located at a bottom **104** of the shell assembly **10** along a height direction G1. The loudspeaker is accommodated in the accommodation cavity, and the protective mesh **13** is connected to the shell assembly **10** and covers the opening mouth of the pressure relief hole **1008**. A projection is carried out along the height direction G1, and a length L1 of a projection of the protective mesh **13** is more than 0.4 times a length L2 of a projection of the shell assembly **10**.

[0081] Specifically, as shown in FIG. 3 and FIG. 10, the accommodation cavity is configured to accommodate the loudspeaker, and the accommodation cavity includes the first accommodation cavity **1001** and a second accommodation cavity **1002** spaced apart. The loudspeaker includes the air-conducting loudspeaker **12** and the bone-conducting loudspeaker **11**, with the air-conducting loudspeaker **12** being accommodated in the second accommodation cavity **1002**, and the bone-conducting loudspeaker **11** being accommodated in the first accommodation cavity **1001**. The specific structure of the shell assembly **10** is described in the content set forth herein and will not be repeated herein in detail. In other embodiments, the accommodation cavity of the earphone **100** may be provided with only the air-conducting loudspeaker **12**, which will not be described in detail herein.

[0082] As shown in FIG. 10 and FIG. 11, the shell assembly **10** is provided with the pressure relief hole **1008** of the air-conducting loudspeaker **12** for pressure relief. The pressure relief hole **1008** is in communication with the accommodation cavity and the external environment, and it can be understood that the pressure relief hole **1008** communicates with the external environment and the second accommodation cavity **1002** for accommodating the air-conducting loudspeaker **12**. Other relative positional relationships between the pressure relief hole **1008** and the accommodation cavity are described above and will not be repeated herein in detail.

[0083] Further, the pressure relief hole **1008** is located at the bottom **104** of the shell assembly **10** along the height direction G1. When the earphone **100** is in a wearing state, the height direction G1 may be approximately parallel to the direction of gravity, and the bottom **104** of the shell assembly

10 is positioned downward along the height direction **G1** or the direction of gravity. Based on this, the opening of the pressure relief hole **1008** being located at the bottom **104** of the shell assembly **10** along the height direction **G1** effectively prevents external adverse factors such as sweat and rain from entering the accommodation cavity through the opening of the pressure relief hole **1008**, which effectively prevents damage to the loudspeaker (specifically, the air-conducting loudspeaker **12** in this embodiment) caused by sweat, rain, and other external adverse factors, thereby significantly enhancing the operational stability of the earphone **100**. Furthermore, the earphone **100** further includes the protective mesh **13** for providing a barrier against external adverse factors such as sweat, rain, and sand. The protective mesh **13** is connected to the shell assembly **10** and covers the opening of the pressure relief hole **1008**. Along a height direction **G1**, the length **L1** of the projection of the protective mesh **13** is more than 0.4 times the length **L2** of the projection of the shell assembly **10**. The length **L1** of the projection is a length of the protective mesh **13** along the air-conducting operating direction **X1**. The length **L1** of the projection of the protective mesh **13** may be within a range of 0.4 to 1 times the length **L2** of the projection of the shell assembly **10**, e.g., 0.4 times, 0.45 times, 0.5 times, 0.55 times, 0.6 times, 0.7 times, 0.8 times, 0.9 times, 1 times, etc. Based on this, the overall dimension and area of the protective mesh **13** can be effectively enhanced, reduces the risk of sweat, rain, and other external adverse factors forming a water film at the protective mesh **13** and blocking the pressure relief hole **1008**, and then effectively ensures that the pressure relief hole **1008** can operate normally, and thus effectively enhances the operational stability of the earphone **100**. It should be understood that since the protective mesh **13** may be curved in shape, an actual length of the protective mesh **13** may be greater than the length **L1** of the projection of the protective mesh **13**. The height direction **G1** is a direction along the bottom **104** of the shell assembly **10** to a top **105** of the shell assembly **10**, as follows.

[0084] Optionally, as shown in FIG. **12**, the protective mesh **13** includes a first portion **13a** that is low along the height direction **G1** and a second portion **13b** that extends from the first portion **13a**. Along the height direction **G1**, as the second portion **13b** gradually extends away from the first portion **13a**, the second portion **13b** is gradually higher than the first portion **13a**. In this manner, the protective mesh **13** may be configured as a curved shape, which may be adapted to the shape of the shell assembly **10**.

[0085] Specifically, the protective mesh **13** includes the first portion **13a** that is low along the height direction **G1** and the second portion **13b** that extends from the first portion **13a**, and as the second portion **13b** gradually extends away from the first portion **13a**, the second portion **13b** is gradually higher than the first portion **13a** along the height direction **G1**. Based on this, the protective mesh **13** is configured as a curved shape, which allows sweat, rain, and other external adverse factors flowing through the second portion **13b** to ultimately drip or slide down to the first portion **13a** before falling off. This effectively prevents the accumulation of sweat, rain, and other external elements in the second portion **13b**, which could otherwise cause clogging. Additionally, positioning the second portion **13b** higher than the first portion **13a** increases the kinetic energy of the sweat, rain, and other external elements, further preventing their accumulation on the protective mesh **13** and avoiding blockage of the pressure relief hole **1008**, thereby enhancing the operational stability of the earphone **100**.

[0086] Optionally, referring further to FIG. **10** and FIG. **11**, the shell assembly **10** is provided with a first recessed portion **1004**. The first recessed portion **1004** is in communication with the pressure relief hole **1008**, and the protective mesh **13** covers the opening of the first recessed portion **1004**. Specifically, the shell assembly **10** is provided with the first recessed portion **1004**, and the first recessed portion **1004** is in communication with the pressure relief hole **1008**. The air-conducting operating direction **X1** is a vibration direction of the air-conducting loudspeaker **12** when it operates, the air-conducting operating direction **X1** is substantially perpendicular to the height direction **G1**. Based on this, the first recessed portion **1004** may be used as an auxiliary pressure relief hole of the air-conducting loudspeaker **12**, which can effectively increase the pressure relief

area of the pressure relief hole, and when the pressure relief hole **1008** is blocked, the pressure of the air-conducting loudspeaker **12** can continue to be relieved through the first recessed portion **1004**, thereby effectively reducing the probability and extent of low-frequency sound loss of the earphone **100**, thereby effectively improving the operational stability of the earphone **100**.

[0087] Optimally, the first recessed portion **1004** extends from the pressure relief hole **1008**. As the first recessed portion **1004** gradually extends away from the pressure relief hole **1008**, the first recessed portion **1004** is gradually higher than the opening of the pressure relief hole **1008** along the height direction **G1**. Specifically, the first recessed portion **1004** is located on a side of the pressure relief hole **1008** along an arrangement direction **X2** that is parallel to the air-conducting operating direction **X1** and perpendicular to the height direction **G1**, and the first recessed portion **1004** extends from the pressure relief hole **1008**. As the first recessed portion **1004** gradually extends away from the pressure relief hole **1008**, the first recessed portion **1004** is gradually higher than the opening of the pressure relief hole **1008**. Based on this, the first recessed portion **1004** can be effectively prevented from being clogged or immersed by external factors such as sweat or rain, thereby effectively enhancing the operational stability of the air-conducting loudspeaker **12**.

[0088] Optionally, referring to FIG. **10**, FIG. **11**, and FIG. **12**, the protective mesh **13** includes the first portion **13a** that is low along the height direction **G1** and the second portion **13b** that extends from the first portion **13a**. As the second portion **13b** gradually extends away from the first portion **13a**, the second portion **13b** is gradually higher than the first portion **13a** along the height direction **G1**. The second portion **13b** covers at least a portion of the opening of the first recessed portion **1004**. Specifically, the protective mesh **13** is configured as a curved shape based on the above-described manner. The second portion **13b** covers at least a portion of the opening of the first recessed portion **1004**. For example, in one embodiment, the first portion **13a** covers a portion of the opening of the first recessed portion **1004**, the second portion **13b** covers the other portion of the opening of the first recessed portion **1004**. Based on this, fully covering the first recessed portion **1004** and the pressure relief hole **1008** with the protective mesh **13** can effectively prevent sweat, rain, and other external factors from clogging the pressure relief hole **1008** and the first recessed portion **1004**. Such an arrangement also effectively prevents other external particulate contaminants from entering the second accommodation cavity through the pressure relief hole **1008** and the first recessed portion **1004**, which could otherwise damage the air-conducting loudspeaker **12**, thereby effectively enhancing the operational stability of the earphone **100**. The protective mesh **13** is configured as a curved shape based on the above-described manner, allowing sweat, rain, and other external contaminants flowing through the second portion **13b** to ultimately drip or slide down to the first portion **13a** before falling off. Such an arrangement effectively prevents the accumulation and crystallization of sweat, rain, and other external contaminants in the second portion **13b**. Additionally, positioning the second portion **13b** higher than the first portion **13a** increases the kinetic energy of the sweat, rain, and other external contaminants, further preventing their accumulation on the protective mesh **13**, which could otherwise clog the pressure relief hole **1008**, thereby effectively enhancing the operational stability of the earphone **100**.

[0089] Optionally, as shown in FIG. **10** and FIG. **11**, the shell assembly **10** includes a support member **1005**, the support member **1005** separating the opening of the pressure relief hole **1008** from the opening of the first recessed portion **1004**, and the support member **1005** is configured to support the protective mesh **13**. Specifically, the support member **1005** is disposed between the pressure relief hole **1008** and the first recessed portion **1004**, and abuts against a centered region of the protective mesh **13**, which can effectively prevent the protective mesh **13** from collapsing due to its large area, thereby ensuring that the protective mesh **13** can function normally and stably, and thereby effectively improving the operational stability of the earphone **100**. Optionally, in some embodiments, the support member **1005** may be integrally molded with the shell assembly **10**. In other embodiments, the support member **1005** may be a separate component that is connected to the shell assembly **10** by a connecting member such as a dispensing adhesive.

[0090] Optionally, as shown in FIG. 10 and FIG. 11, the shell assembly 10 is provided with a sound receiving hole 1007, the sound receiving hole 1007 is disposed adjacent to the opening of the pressure relief hole 1008, and the protective mesh 13 also covers the sound receiving hole 1007. Specifically, the sound receiving hole 1007 of the earphone 100 that is configured to receive sound information (e.g., the user's voice enters the microphone of the earphone 100 via the sound receiving hole 1007) is disposed adjacent to the pressure relief hole 1008, and the protective mesh 13 also covers the sound receiving hole 1007. The protective mesh 13 effectively protects the sound receiving hole 1007 from erosion by sweat, rain, and other external adverse factors. Along the height direction G1, the length L1 of the projection of the protective mesh 13 is more than 0.4 times the length L2 of the projection of the shell assembly 10. Based on this, it can be ensured that the length of the protective mesh 13 is long enough so that the sound receiving hole 1007 and the pressure relief hole 1008 are staggered from each other as much as possible, thereby effectively ensuring the sound quality of the earphone 100.

[0091] Optionally, as shown in FIG. 10 and FIG. 11, the opening of the sound receiving hole 1007 is higher than the opening of the pressure relief hole 1008 along the height direction G1. By setting the sound receiving hole 1007 higher than the bottom of the shell assembly 10 along the height direction G1, the sound receiving hole 1007 can be effectively prevented from being eroded by sweat, rain, and other undesirable external factors.

[0092] Optionally, an area of the protective mesh 13 is more than 1.5 times an area of the opening of the pressure relief hole 1008. For example, the area of the protective mesh 13 may be more than 1.5 times, 1.8 times, 2 times, 2.5 times, 3 times, 4 times, 5 times, 6 times, 7 times, 8 times, 9 times, 10 times, etc., the area of the opening of the pressure relief hole 1008. Based on this, it can ensure that the area of the protective mesh 13 is sufficiently large, thereby reducing the risk of sweat, rain, and other external contaminants forming a water film on the protective mesh 13 and blocking the pressure relief hole 1008, which ensures that the pressure relief hole 1008 can function properly, thereby significantly enhancing the operational stability of the earphone 100. Further, based on the above setting, the area of the protective mesh 13 is set to be sufficiently large relative to the pressure relief hole 1008, thereby providing more effective protection for the first recessed portion 1004, the sound receiving hole 1007, and the pressure relief hole 1008 together, improving the protection efficiency. At the same time, placing multiple structures such as the first recessed portion 1004, the sound receiving hole 1007, and the pressure relief hole 1008 at the bottom 104 of the shell assembly 10 allows for full utilization of the structural space of the shell assembly 10.

[0093] Optionally, as shown in FIG. 10 and FIG. 11, the protective mesh 13 includes the first portion 13a that is low along the height direction G1 and the second portion 13b that extends from the first portion 13a. As the second portion 13b gradually extends away from the first portion 13a, the second portion 13b is gradually higher than the first portion 13a along the height direction G1; and the second portion 13b covers the opening of the sound receiving hole 1007. Specifically, the protective mesh 13 is configured as a curved shape based on the above manner. The second portion 13b that is higher than the first portion 13a covers the opening of the sound receiving hole 1007. For example, in one embodiment, the first portion 13a covers a portion of the opening of the first recessed portion 1004, and the second portion 13b covers the opening of the sound receiving hole 1007 and the other portion of the first recessed portion 1004. Based on this, the sound receiving hole 1007, the first recessed portion 1004, and the pressure relief hole 1008 are completely covered by the protective mesh 13, so that the sound receiving hole 1007, the pressure relief hole 1008, and the first recessed portion 1004 can be effectively prevented from being clogged by sweat, rain, and other external factors, as well as effectively preventing other external particulate contaminants from entering the second accommodation cavity and causing damage to the air-conducting loudspeaker 12, thereby enhancing the operational stability of the earphone 100.

[0094] Optionally, as shown in FIG. 10 and FIG. 11, the sound receiving hole 1007 is formed on the support wall 1006, and the protective mesh 13 abuts against the support wall 1006. Specifically,

the support wall **1006** serves as a portion of the shell assembly **10**. The sound receiving hole **1007** is formed on the support wall **1006**, and the support wall **1006** fits with the protective mesh **13** and supports the protective mesh **13**, thereby effectively preventing the protective mesh **13** from collapsing under an external force, affecting the normal operation of the earphone **100**, and thereby effectively enhancing the operational stability of the earphone **100**. Optionally, the support wall **1006** may be a portion of the bottom of the shell assembly **10** that is recessed along the height direction **G1**. Along the height direction **G1**, the first recessed portion **1004** is located in a different plane from the support wall **1006**. For example, along the height direction **G1**, the first recessed portion **1004** is higher than the support wall **1006**, based on this, along the height direction **G1**, the sound receiving hole **1007** is provided at a position lower than the first recessed portion **1004**, effectively preventing the first recessed portion **1004** from affecting the sound receiving operation of the sound receiving hole **1007** when the first recessed portion **1004** performs a pressure relief function.

[0095] Optionally, as shown in FIG. **9** and FIG. **10**, the shell assembly **10** is provided with a second recessed portion **1009**, and the protective mesh **13** is provided in the second recessed portion **1009**. Specifically, the second recessed portion **1009** may be formed by an overall recess in a specific area of the bottom of the shell assembly **10** along the height direction **G1**, and the second recessed portion **1009** is configured to accommodate the protective mesh **13**. An outer peripheral rim of the second recessed portion **1009** may be further recessed along the height direction **G1**, so that an outer peripheral rim of the protective mesh **13** abuts against and is fixedly connected to the outer peripheral rim of the second recessed portion **1009**, e.g., the protective mesh **13** is fixedly connected to the second recessed portion **1009** by dispensing adhesive on the outer peripheral rim of the second recessed portion **1009**. The second recessed portion **1009** accommodates and fixes the protective mesh **13**. Based on this, an outer surface of the protective mesh **13** transitions smoothly with an outer surface of the shell assembly **10**, thereby effectively enhancing the rounded and smooth shape of an outer contour of the earphone **100**.

[0096] Optionally, as shown in FIG. **10** and FIG. **11**, the protective mesh **13** includes a steel mesh **131**, a gauze mesh **132**, and a waterproof membrane **133**. The side of the steel mesh **131** facing the opening of the pressure relief hole **1008** is provided with the gauze mesh **132**, and the side of the gauze mesh **132** facing the opening of the pressure relief hole **1008** is provided with the waterproof membrane **133**. Specifically, in one embodiment, the protective mesh **13** includes the steel mesh **131**, the gauze mesh **132**, and the waterproof membrane **133**. The side of the steel mesh **131** facing the opening of the pressure relief hole **1008**, i.e., the side of the steel mesh **131** near the pressure relief hole **1008** (also referred to as a side near the shell assembly **10**) is provided with the gauze mesh **132**. Based on this, the gauze mesh **132** can effectively prevent external particulate contaminants from entering the pressure relief hole **1008**, thereby protecting the air-conducting loudspeaker **12** from damage. The side of the gauze mesh **132** facing the opening of the pressure relief hole **1008**, i.e., the side of the gauze mesh **132** near the pressure relief hole **1008** (also referred to as a side near the shell assembly **10**) is provided with the waterproof membrane **133**. Based on this, the waterproof membrane **133** can effectively prevent external undesirable factors such as sweat, and rain from entering into the second accommodation cavity **1002** via the pressure relief hole **1008**, thereby effectively protecting the air-conducting loudspeaker **12**, and improving the operational stability of the earphone **100**. Optionally, in one embodiment, the side of the gauze mesh **132** facing the opening of the pressure relief hole **1008** is provided with at least two layers of the waterproof membrane **133**, thereby effectively improving the protective performance of the protective mesh **13**.

[0097] Optionally, the mesh holes of the steel mesh **131** are smaller than the mesh holes of the gauze mesh **132**. Based on this, it can effectively reduce the wind noise of the pressure relief hole **1008**, thereby effectively improving the sound quality of the earphone **100**.

[0098] Optionally, as shown in FIG. **3**, FIG. **13**, FIG. **14**, and FIG. **15**, the present disclosure also

presents a second embodiment of the pressure relief hole **1008** protection scheme. The second embodiment of the pressure relief hole **1008** protection scheme applies to the earphone **100** provided with both the bone-conducting loudspeaker **11** and the air-conducting loudspeaker **12**, and it may also apply to the earphone **100** provided with the air-conducting loudspeaker **12** alone. Herein, the second embodiment of the pressure relief hole **1008** protection scheme is mainly described in terms of the earphone **100** provided with both the air-conducting loudspeaker **12** and the bone-conducting loudspeaker **11**. The second embodiment of the pressure relief hole **1008** protection scheme is described as follows.

[0099] The loudspeaker assembly **1** includes the shell assembly **10**, the loudspeaker, and a first annular protrusion **20**. The shell assembly **10** is provided with an accommodation cavity, the shell assembly **10** is provided with the pressure relief hole **1008** for communicating with the accommodation cavity and the external environment, and the opening of the pressure relief hole **1008** is provided at the bottom **104** of the shell assembly **10** along the height direction **G1**. The loudspeaker is accommodated in the accommodation cavity; the first annular protrusion **20** is disposed around the opening of the pressure relief hole **1008** and extends downwardly from the bottom **104** of the shell assembly **10**.

[0100] Specifically, as illustrated in FIG. 3, the accommodation cavity is used to accommodate the loudspeaker, and the accommodation cavity includes the first accommodation cavity **1001** and the second accommodation cavity **1002** spaced apart, the loudspeaker includes the air-conducting loudspeaker **12** and the bone-conducting loudspeaker **11**, with the air-conducting loudspeaker **12** being accommodated in the second accommodation cavity **1002** and the bone-conducting loudspeaker **11** being accommodated in the first accommodation cavity **1001**. Specific structures of the shell assembly **10** are described above, and will not be repeated herein in detail. In other embodiments, the accommodation cavity of the earphone **100** may be provided with only the air-conducting loudspeaker **12**, which will not be described in detail herein. In one embodiment, the shell assembly **10** is provided with the pressure relief hole **1008** for a loudspeaker (e.g., the air-conducting loudspeaker **12**) to relieve pressure, the pressure relief hole **1008** being in communication with the accommodation cavity (e.g., the second accommodation cavity **1002**) and the external environment. Other relative positional relationships between the pressure relief hole **1008** and the accommodation cavity are described in the foregoing and will not be discussed in detail herein. Distinguishing from the above embodiments, in the present embodiment, the loudspeaker assembly **1** includes the first annular protrusion **20**. The first annular protrusion **20** is disposed around an outer circumferential side of the opening of the pressure relief hole **1008** and extends downwardly from the bottom **104** of the shell assembly **10**. The first annular protrusion **20** extending downwardly from the bottom **104** of the shell assembly **10** can be understood that the first annular protrusion **20** protrudes from a bottom surface of the shell assembly **10** along the height direction **G1**. Based on this, the first annular protrusion **20** can effectively prevent external undesirable factors such as sweat, and rain from flowing into the pressure relief hole **1008** and causing damage to the loudspeaker in the accommodation cavity, thereby effectively enhancing the operational stability of the earphone **100**. The height direction **G1** mentioned here and hereinafter is a direction along the bottom **104** of the shell assembly **10** to the top **105** of the shell assembly **10**. Optionally, in some embodiments, the first annular protrusion **20** may be disposed around both the opening of the pressure relief hole **1008** and the outer circumferential side of the sound receiving hole **1007**, which effectively prevents external undesirable factors such as sweat, and rain from flowing into the pressure relief hole **1008** and the sound receiving hole **1007** and causing damage to the loudspeaker in the accommodation cavity, thereby effectively improving the operational stability of the earphone **100**.

[0101] Optionally, the first annular protrusion **20** is a metal ring or plastic ring, which is bonded to the bottom **104** of the shell assembly **10**. Specifically, in some embodiments, the first annular protrusion **20** may be a structural member such as a metal ring or a plastic ring, which is fixedly

connected to the shell assembly **10** using bonding agents like adhesive.

[0102] Optionally, the first annular protrusion **20** is an integrally molded structure with a portion of the shell assembly **10**. Specifically, in some embodiments, a portion of the bottom of the shell assembly **10** that is located on an outer circumferential side of the opening of the pressure relief hole **1008** protrudes along the height direction away from the pressure relief hole **1008** to form the first annular protrusion **20**. Based on this, the first annular protrusion **20** and the shell assembly **10** form an integrally-molded structure. In embodiments where the shell assembly **10** includes the shell **101**, the shell **102**, and the shell **103**, the first annular protrusion **20** may be integrally molded with the shell **101**, or the first annular protrusion **20** may be integrally molded with the shell **102**; or, a portion of the first annular protrusion **20** may be integrally molded with the shell **101**, and the other portion of the first annular protrusion **20** may be integrally molded with the shell **102**. With the integrally-molded structure, the preparation process of the first annular protrusion **20** can be simplified, the structural strength of the first annular protrusion **20** can be strengthened, and the cost of production can be reduced.

[0103] Optionally, as shown in FIG. **14**, the earphone **100** further includes the protective mesh **13**, the protective mesh **13** covering the opening of the pressure relief hole **1008**. Along the height direction **G1**, the protective mesh **13** is higher than a bottom surface **21** of the first annular protrusion **20**. Specifically, the protective mesh **13** is used to cover the opening of the pressure relief hole **1008** to prevent external undesirable factors such as sweat, and rain from entering the pressure relief hole **1008**. Further, in some embodiments, the protective mesh **13** is higher than the bottom surface **21** of the first annular protrusion **20** along the height direction **G1**, i.e., there is a difference in height between a bottom surface **134** of the protective mesh **13** and the bottom surface **21** of the first annular protrusion **20** to allow the protective mesh **13** to be fully submerged in the region enclosed by the first annular protrusion **20** while the protective mesh **13** still exists a certain height difference. Such an arrangement can reduce the direct contact between external environmental factors such as rain, sweat, and the protective mesh **13** to some extent, thereby decreasing the likelihood of the pressure relief hole **1008** being infiltrated by external environmental factors such as rain, and sweat.

[0104] Optionally, as shown in FIG. **15**, in other embodiments, the earphone **100** includes the protective mesh **13**, and an inner side surface of the first annular protrusion **20** includes an annular support surface **24**, the annular support surface **24** is connected to the protective mesh **13**. Along the height direction **G1**, the bottom surface **134** of the protective mesh **13** is flush with the bottom surface **21** of the first annular protrusion **20**. Specifically, in one embodiment, the first annular protrusion **20** includes a first portion **22** and a second portion **23** connected to the first portion **22** along the height direction **G1**, the first portion **22** being connected to the bottom **104** of the shell assembly **10**; a cross-sectional dimension of an internal channel of the first portion **22** is smaller than a cross-sectional dimension of an internal channel of the second portion **23**, and the annular support surface **24** is formed at a transition between the first portion **22** and the second portion **23**. The annular support surface **24** protrudes from the pressure relief hole **1008**, the annular support surface **24** is used to support the protective mesh **13** and is fixedly connected to the protective mesh **13**, enabling the bottom surface **134** of the protective mesh **13** to flush with the bottom surface **21** of the first annular protrusion **20**, i.e., the difference in height between the bottom surface **134** of the protective mesh **13** and the bottom surface **21** of the first annular protrusion **20** is zero. Based on this, it can effectively prevent external factors such as sweat, and rainwater from accumulating in the space between the bottom surface **134** of the protective mesh **13** and the bottom surface **21** of the first annular protrusion **20**, thereby preventing the pressure relief hole **1008** from being clogged, thereby enhancing the operational stability of the earphone **100**.

[0105] Optionally, as shown in FIG. **15**, an outer rim of the annular support surface **24** is provided with an annular recess **25**, and a bonding agent is disposed inside the annular recess **25**. Specifically, the outer rim of the annular support surface **24** is the position where the annular

support surface **24** is proximate to the second portion **23**, and the annular support surface **24** is formed with an annular recess **25** at the position. An adhesive such as glue, etc. is disposed in the annular recess **25**. The protective mesh **13** is connected to the annular support surface **24** by the adhesive, and can be effectively fixedly connected to the annular support surface **24**.

[0106] Optionally, as shown in FIG. **14** and FIG. **15**, the type of the protective mesh **13** is similar to that of the embodiments described above, i.e., the protective mesh **13** may include the steel mesh **131**, the gauze mesh **132**, and the waterproofing membrane **133**, with the side of the steel mesh **131** facing the opening of the pressure relief hole **1008** being provided with the gauze mesh **132**, and the side of the gauze mesh **132** facing the opening of the pressure relief hole **1008** being provided with the waterproof membrane **133**. In other words, in one embodiment, the protective mesh **13** includes the steel mesh **131**, the gauze mesh **132**, and the waterproof membrane **133**. The side of the steel mesh **131** facing the opening of the pressure relief hole **1008**, i.e., the side of the steel mesh **131** near the pressure relief hole **1008** (also referred to as a side near the shell assembly **10**) is provided with the gauze mesh **132**. Based on this, the gauze mesh **132** can effectively prevent the outside particulate contaminants from entering the pressure relief hole **1008**, thereby protecting the air-conducting loudspeaker **12** from damage. The side of the gauze mesh **132** facing the opening of the pressure relief hole **1008**, i.e., the side of the gauze mesh **132** near the pressure relief hole **1008** (also referred to as a side near the shell assembly **10**) is provided with the waterproof membrane **133**. Based on this, the waterproof membrane **133** can effectively prevent external undesirable factors such as sweat, and rain from entering into the second accommodation cavity **1002** via the pressure relief hole **1008**, thereby effectively protecting the air-conducting loudspeaker **12** and improving the operational stability of the earphone **100**. Optimally, in one embodiment, a side of the gauze mesh **132** facing the opening of the pressure relief hole **1008** is provided with at least two layers of the waterproof membrane **133**. Based on this, the protective performance of the protective mesh **13** can be effectively improved.

[0107] Optionally, the mesh holes of the steel mesh **131** are smaller than the mesh holes of the gauze mesh **132**. Based on this, it can effectively reduce the wind noise of the pressure relief hole **1008**, thereby effectively improving the sound quality of the earphone **100**.

[0108] The foregoing is only a part of the embodiments of the present disclosure, and is not intended to limit the scope of protection of the present disclosure, and any equivalent device or equivalent process transformations utilizing the contents of the present disclosure and the accompanying drawings, or applying them directly or indirectly in other related technical fields, are similarly included in the scope of patent protection of the present disclosure.

[0109] The above is only an example of the present disclosure, and is not intended to limit the scope of the present disclosure, and any equivalent structure or equivalent process transformations utilizing the contents of the present disclosure and the accompanying drawings, or directly or indirectly applying them in other related technical fields, are included in the scope of the present disclosure.

Claims

1. A loudspeaker assembly, comprising: a shell assembly provided with a first accommodation cavity and a second accommodation cavity; a bone-conducting loudspeaker accommodated in the first accommodation cavity; and an air-conducting loudspeaker accommodated in the second accommodation cavity, the air-conducting loudspeaker including: a magnetic circuit system including a magnet and a magnetic conduction assembly, the magnet being fixed on the magnetic conduction assembly, the magnetic conduction assembly being fixed in the second accommodation cavity, and the magnetic conduction assembly being provided with a weight reduction structure, wherein the magnetic conduction assembly includes a magnetic conduction shield and a magnetic conduction plate, the magnet being fixed in the magnetic conduction shield, the magnetic

conduction plate being fixed to the magnet, the weight reduction structure being arranged on at least one of the magnetic conduction shield or the magnetic conduction plate.

2. The loudspeaker assembly of claim 1, wherein the weight reduction structure includes at least one of a groove or a through hole arranged at the magnetic conduction assembly.

3. The loudspeaker assembly of claim 1, wherein a magnetic induction intensity at a position of the magnetic conduction assembly where the weight reduction structure is arranged is less than a magnetic induction intensity at other positions of the magnetic conduction assembly.

4. The loudspeaker assembly of claim 1, wherein there is one or more weight reduction structures on the magnetic conduction shield and the magnetic conduction plate of the magnetic conduction assembly, and a plurality of weight reduction structures are spaced apart.

5. The loudspeaker assembly of claim 4, wherein the magnetic conduction shield includes a bottom wall and an annular side wall extending from the bottom wall, and the weight reduction structure is arranged at the bottom wall.

6. The loudspeaker assembly of claim 4, wherein a total area of the weight reduction structure on the magnetic conduction shield accounts for less than or equal to 50% of an area of a bottom wall of the magnetic conduction shield, and a total area of the weight reduction structure on the magnetic conduction plate accounts for less than or equal to 50% of an area of the magnetic conduction plate.

7. The loudspeaker assembly of claim 4, wherein a total area of the weight reduction structure on the magnetic conduction shield accounts for less than or equal to 30% of an area of a bottom wall of the magnetic conduction shield, and a total area of the weight reduction structure on the magnetic conduction plate accounts for less than or equal to 30% of an area of the magnetic conduction plate.

8. The loudspeaker assembly of claim 6, wherein a planar shape of the weight reduction structure on the magnetic conduction shield is the same as a planar shape of the bottom wall of the magnetic conduction shield and is reduced in equal proportion relative to the planar shape of the bottom wall of the magnetic conduction shield; and a planar shape of the weight reduction structure on the magnetic conduction plate is the same as a planar shape of the magnetic conduction plate and is reduced in equal proportion relative to the planar shape of the magnetic conduction plate.

9. The loudspeaker assembly of claim 5, wherein the weight reduction structure is centrally arranged at the bottom wall.

10. The loudspeaker assembly of claim 4, wherein the weight reduction structure is centrally arranged at the magnetic conduction plate.

11. The loudspeaker assembly of claim 9, wherein the weight reduction structure is a through hole.

12. The loudspeaker assembly of claim 1, wherein the magnet is provided with the weight reduction structure, and the weight reduction structure is a groove or a through hole.

13. The loudspeaker assembly of claim 12, wherein a magnetic induction intensity at a position of the magnet where the weight reduction structure is arranged is less than a magnetic induction intensity at other positions of the magnet.

14. The loudspeaker assembly of claim 12, wherein the weight reduction structure on the magnet is located at a center of the magnet, and a planar shape of the weight reduction structure on the magnet is the same as a planar shape of the magnet and is reduced in equal proportion relative to the planar shape of the magnet.

15. The loudspeaker assembly of claim 12, wherein a planar area of the weight reduction structure on the magnet accounts for less than or equal to 50% of a planar area of the magnet.

16. The loudspeaker assembly of claim 1, wherein the air-conducting loudspeaker further includes a fixing assembly, the fixing assembly is fixed to the magnetic conduction assembly, and the fixing assembly is fixedly connected to the shell assembly.

17. The loudspeaker assembly of claim 16, wherein the air-conducting loudspeaker further includes a vibration diaphragm and a voice coil, a rim of the vibration diaphragm is connected to a periphery

of the fixing assembly, and the voice coil is connected to the vibration diaphragm.

18. The loudspeaker assembly of claim 1, wherein the loudspeaker assembly is provided with a pressure relief hole for communicating with the second accommodation cavity and external environment, and the pressure relief hole extends toward a side where the first accommodation cavity is located.

19. A wearable electronic device, comprising a loudspeaker assembly, wherein the loudspeaker assembly includes: a shell assembly provided with a first accommodation cavity and a second accommodation cavity; a bone-conducting loudspeaker accommodated in the first accommodation cavity; and an air-conducting loudspeaker accommodated in the second accommodation cavity, the air-conducting loudspeaker including: a magnetic circuit system including a magnet and a magnetic conduction assembly, the magnet being fixed on the magnetic conduction assembly, the magnetic conduction assembly being fixed in the second accommodation cavity, and the magnetic conduction assembly being provided with a weight reduction structure, wherein the magnetic conduction assembly includes a magnetic conduction shield and a magnetic conduction plate, the magnet being fixed in the magnetic conduction shield, the magnetic conduction plate being fixed to the magnet, the weight reduction structure being arranged on at least one of the magnetic conduction shield or the magnetic conduction plate.

20. The wearable electronic device of claim 19, wherein the wearable electronic device is earphones, glasses, an AR device, or a VR device.
